

BOULDER, NV, USA

WMO: 720741

Lat: 35.947N

Lon: 114.861W

Elev: 671

StdP: 93.52

Time Zone: -8.00 (NAP)

Period: 10-19

WBAN: 00269

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-0.2	1.1	-19.8	0.7	13.4	-17.6	0.9	13.9	11.8	10.7	10.9	9.8	1.7	340	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	42.5	18.2	41.2	18.2	40.0	18.1	22.0	33.6	21.4	33.2	20.9	32.8	4.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.9	25.8	17.7	13.8	26.8	17.0	13.2	28.2	67.6	33.7	65.5	32.8	63.4	32.3	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	10.4	8.9	-2.8	44.6	1.4	0.9	-3.8	45.2	-4.6	45.7	-5.4	46.2	-6.4	46.8		
(5)			-5.9	23.1	1.4	0.5	-6.9	23.5	-7.7	23.8	-8.4	24.0	-9.4	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.5	9.6	11.5	16.0	19.8	23.3	30.5	32.6	31.7	27.7	20.5	13.7	8.7		
	DBStd	9.05	3.25	3.96	3.66	3.91	4.19	3.19	2.49	2.46	3.35	3.71	3.89	3.11		
	HDD10.0	149	46	28	2	1	0	0	0	0	0	1	11	62		
	HDD18.3	1057	272	191	91	29	9	0	0	0	0	21	145	299		
	CDD10.0	3982	32	72	187	294	413	614	699	671	532	326	121	21		
	CDD18.3	1847	0	1	17	73	163	364	441	413	282	88	5	0		
	CDH23.3	27485	0	15	229	854	2124	5942	7257	6442	3697	860	64	0		
	CDH26.7	17038	0	0	54	323	1074	3964	4961	4228	2134	296	3	0		
(14) Wind	WSAvg	3.2	2.5	3.0	3.5	4.0	3.9	3.8	3.4	3.0	2.8	2.8	2.6	2.5		
(15) Precipitation	PrecAvg	152	15	18	19	8	5	3	11	21	14	7	14	11		
	PrecMax	339	73	80	128	77	35	18	53	129	95	57	67	62		
	PrecMin	35	0	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	69	19	20	25	14	7	5	14	28	22	12	18	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	25.2	31.0	34.2	38.0	44.8	44.0	42.5	40.1	33.9	27.2	20.1		
		MCWB	9.9	11.3	13.2	14.0	15.3	18.1	19.1	18.5	18.5	16.3	12.6	10.1		
	2%	DB	18.8	22.9	27.9	32.5	36.4	42.5	42.6	41.4	38.0	32.2	25.1	17.9		
		MCWB	8.4	10.3	12.1	13.5	14.7	17.6	18.6	18.4	17.9	15.2	11.5	8.9		
	5%	DB	17.4	21.3	26.1	30.2	34.1	41.1	41.3	40.1	37.3	29.9	23.0	16.3		
		MCWB	7.7	9.8	10.9	12.8	13.8	17.1	18.9	18.5	17.5	14.3	10.7	7.8		
	10%	DB	15.9	18.9	23.8	27.9	32.5	39.1	39.9	38.9	36.0	27.8	21.2	14.7		
		MCWB	7.0	8.8	10.2	11.8	13.5	16.2	18.8	18.5	17.2	13.4	10.1	7.4		
	0.4%	WB	11.7	13.6	13.8	15.1	16.4	20.5	22.7	22.8	21.9	19.1	13.8	12.6		
		MCDB	15.9	21.1	26.9	31.5	33.2	38.8	34.4	33.5	29.8	25.4	24.5	15.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	10.6	11.7	12.8	14.1	15.5	19.1	22.0	22.0	20.8	17.4	12.6	11.2		
		MCDB	14.5	19.0	25.2	29.0	32.8	37.3	33.9	33.7	30.4	27.9	23.3	15.1		
	5%	WB	9.6	10.6	11.9	13.3	14.8	18.2	21.5	21.4	20.1	16.2	11.5	10.0		
		MCDB	14.4	18.1	23.4	27.3	31.6	37.4	33.7	33.1	30.6	25.4	21.8	14.5		
	10%	WB	8.4	9.6	11.0	12.5	14.0	17.2	21.0	20.8	19.4	15.1	10.4	8.6		
		MCDB	13.9	18.5	22.4	26.0	30.0	36.6	33.5	32.5	30.6	24.5	20.1	12.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	11.6	12.7	13.2	13.5	15.5	13.6	13.6	13.6	12.7	12.1	10.0		
		MCDBR	12.6	14.6	15.8	16.5	16.0	16.5	15.3	15.4	15.1	14.5	14.3	12.3		
		MCWBR	6.5	6.5	6.2	5.9	5.2	4.6	4.1	4.5	5.1	5.4	6.4	6.5		
	5% WB	MCDBR	9.7	12.2	14.0	14.4	14.7	14.6	11.2	11.2	12.1	11.0	13.2	9.2		
		MCWBR	5.4	5.7	5.7	5.5	4.8	4.4	3.2	3.9	4.4	4.0	6.2	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.262	0.267	0.280	0.301	0.318	0.316	0.355	0.347	0.313	0.288	0.265	0.262		
	taud		2.632	2.637	2.570	2.483	2.461	2.476	2.457	2.470	2.548	2.607	2.639	2.641		
	Ebn at Noon		960	991	997	982	963	960	920	924	947	951	948	940		
	Edh at Noon		71	79	93	107	112	110	111	108	94	81	70	66		
(44) All-Sky Solar Radiation	RadAvg		3.10	4.06	5.56	6.97	7.92	8.56	7.55	7.00	6.19	4.66	3.48	2.74		
	RadStd		0.24	0.34	0.27	0.29	0.36	0.32	0.38	0.34	0.25	0.26	0.14	0.20		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.49	N/A	-1.57	N/A	N/A	N/A	-117	+189	+81

Nomenclature: See separate page